Multiplication

Equal groups: count the groups, then how many in each.

3 groups of 4



$4 + 4 + 4 = 12$

$3 \times 4 = 12$ say “3 groups of 4”

An array: rows × how many in each row



$2 \times 5 = 10$ 2 rows of 5

Turn it around: same dots, same answer



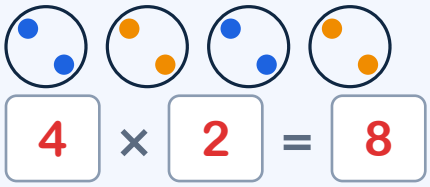
$5 \times 2 = 10$ 5 columns of 2

Count by to multiply



$4 \times 5 = 20$

Your turn!



For teachers

Goal: multiplication means equal groups (3.OA.1). Read 3 × 4 as “3 groups of 4.” Build on packet 21: 4 + 4 + 4 is 3 × 4. Turnarounds give the same answer. Learn ×2, ×5 and ×10 by counting by.

Multiplication: Practice

Count the groups. Count how many in each. Multiply.

Part 1 Groups of Write the groups, then the multiplication.

1) 

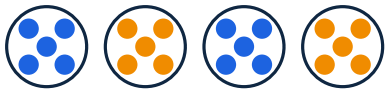
3 groups of 4

3 × 4 = 12

2) 

5 groups of 2

5 × 2 = 10

3) 

4 groups of 5

4 × 5 = 20

4) 

2 groups of 3

2 × 3 = 6

5) 

4 groups of 3

4 × 3 = 12

6) 

5 groups of 5

5 × 5 = 25

Part 2 Add, then multiply The same groups, written two ways.

7) 5 + 5 + 5 = 15

3 × 5 = 15

8) 2 + 2 + 2 + 2 = 8

4 × 2 = 8

9) 10 + 10 + 10 = 30

3 × 10 = 30

10) 3 + 3 + 3 + 3 + 3 = 15

5 × 3 = 15

Multiplication: Practice (continued)

Part 2 Add, then multiply (continued)

11) $4 + 4 = 8$

$2 \times 4 = 8$

12) $6 + 6 + 6 = 18$

$3 \times 6 = 18$

13) $10 + 10 = 20$

$2 \times 10 = 20$

14) $2 + 2 + 2 + 2 + 2 + 2 = 12$

$6 \times 2 = 12$

Part 3 Arrays Rows times how many in each row.

15) 

2 rows of 4

$2 \times 4 = 8$

16) 

3 rows of 5

$3 \times 5 = 15$

17) 

4 rows of 4

$4 \times 4 = 16$

18) 

5 rows of 2

$5 \times 2 = 10$

19) 

2 rows of 6

$2 \times 6 = 12$

20) 

3 rows of 3

$3 \times 3 = 9$

Multiplication: Practice (continued)

Part 4 Turn it around Read the rows, then the columns.

21)



rows
4 × 3 = 12
columns
3 × 4 = 12

22)



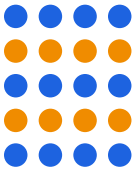
rows
2 × 7 = 14
columns
7 × 2 = 14

23)



rows
3 × 6 = 18
columns
6 × 3 = 18

24)



rows
5 × 4 = 20
columns
4 × 5 = 20

Part 5 Draw it

25) **Draw 2 groups of 5.**

Draw it



2 × 5 = 10

26) **Draw 3 groups of 2.**

Draw it



3 × 2 = 6

27) **Draw 5 groups of 3.**

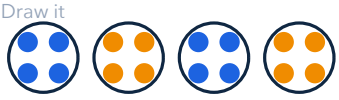
Draw it



5 × 3 = 15

28) **Draw 4 groups of 4.**

Draw it



4 × 4 = 16

Part 6 Count by Fill in the count. The last number is the answer.

29) Count by 5s, 6 times: 5, 10, 15, 20, 25, 30

6 × 5 = 30

Multiplication: Practice (continued)

Part 6 Count by (continued)

30) Count by 10s, 4 times: 10, 20, 30, 40

$4 \times 10 = 40$

31) Count by 2s, 8 times: 2, 4, 6, 8, 10, 12, 14, 16

$8 \times 2 = 16$

32) Count by 5s, 7 times: 5, 10, 15, 20, 25, 30, 35

$7 \times 5 = 35$

33) Count by 10s, 7 times: 10, 20, 30, 40, 50, 60, 70

$7 \times 10 = 70$

34) Count by 2s, 6 times: 2, 4, 6, 8, 10, 12

$6 \times 2 = 12$

Part 7 Times 2 No help this time. Times 2 is a double.

35) $2 \times 6 = 12$

36) $7 \times 2 = 14$

37) $2 \times 9 = 18$

38) $4 \times 2 = 8$

39) $2 \times 3 = 6$

40) $8 \times 2 = 16$

Multiplication: Practice (continued)

Part 7 Times 2 (continued)

41) $2 \times 7 = 14$

42) $6 \times 2 = 12$

43) $2 \times 4 = 8$

Part 8 Times 10 No help this time. Count by 10s.

44) $10 \times 4 = 40$

45) $6 \times 10 = 60$

46) $10 \times 8 = 80$

47) $3 \times 10 = 30$

48) $10 \times 7 = 70$

49) $9 \times 10 = 90$

50) $10 \times 2 = 20$

51) $5 \times 10 = 50$

52) $10 \times 6 = 60$

Multiplication: Practice (continued)

Part 9 Times 5 No help this time. Times 5 is half of times 10.

53) $5 \times 3 = 15$	54) $8 \times 5 = 40$	55) $5 \times 6 = 30$
56) $4 \times 5 = 20$	57) $5 \times 9 = 45$	58) $7 \times 5 = 35$
59) $5 \times 2 = 10$	60) $6 \times 5 = 30$	61) $5 \times 8 = 40$

Part 10 Mixed No help this time.

62) $10 \times 10 = 100$	63) $9 \times 2 = 18$	64) $5 \times 7 = 35$

Multiplication: Practice (continued)

Part 10 Mixed (continued)

65) $10 \times 3 = 30$

66) $2 \times 2 = 4$

67) $5 \times 5 = 25$

68) $7 \times 10 = 70$

69) $3 \times 2 = 6$

70) $5 \times 4 = 20$

71) $10 \times 9 = 90$

72) $2 \times 8 = 16$

73) $9 \times 5 = 45$

Part 11 Groups of

74) 4 groups of 6 = 24

75) 2 groups of 7 = 14

Multiplication: Practice (continued)

Part 11 Groups of (continued)

76) 4 groups of 3 = 12

77) 2 groups of 2 = 4

78) 4 groups of 4 = 16

79) 2 groups of 8 = 16

80) 4 groups of 10 = 40

81) 5 groups of 5 = 25

82) 2 groups of 10 = 20

83) 3 groups of 6 = 18

84) 5 groups of 6 = 30

85) 2 groups of 9 = 18

86) 3 groups of 4 = 12

87) 2 groups of 6 = 12

Multiplication: Practice (continued)

Part 11 Groups of (continued)

88) 3 groups of 3 = 9

89) 2 groups of 3 = 6

90) 3 groups of 2 = 6

91) 5 groups of 3 = 15

92) 2 groups of 5 = 10

93) 5 groups of 10 = 50

94) 4 groups of 2 = 8

95) 5 groups of 7 = 35

96) 3 groups of 5 = 15

97) 5 groups of 2 = 10

Multiplication: Practice (continued)

Part 12 Times 2, 5 and 10

98) $1 \times 2 =$ **2**

99) $2 \times 1 =$ **2**

100) $1 \times 5 =$ **5**

101) $5 \times 1 =$ **5**

102) $1 \times 10 =$ **10**

103) $10 \times 1 =$ **10**

Part 13 Remember? Problems from earlier packets.

104) 36 cm + 47 cm = **83** cm

105) quarter to 2 = **1:45**

106) 2 quarters + 2 dimes = **70¢**

107) $900 - 418 =$ **482**

108) 3 thirds = **1** whole

109) $57 + 38 =$ **95**

Multiplication: Practice (continued)

Part 14 Word problems Find the groups and how many in each.

110) Mia has 4 bags. Each bag has 5 apples. How many apples does she have?

Number sentence $4 \times 5 = 20$

Answer **20** apples

Draw or show your work

111) A box of crayons has 3 rows with 10 crayons in each row. How many crayons are in the box?

Number sentence $3 \times 10 = 30$

Answer **30** crayons

Draw or show your work

112) 6 kids each wear 2 shoes. How many shoes is that?

Number sentence $6 \times 2 = 12$

Answer **12** shoes

Draw or show your work

113) Ben buys 7 packs of gum. Each pack has 5 pieces. How many pieces does he have?

Number sentence $7 \times 5 = 35$

Answer **35** pieces

Draw or show your work

114) Challenge: A spider has 8 legs and an ant has 6. How many legs do 2 spiders and 1 ant have?

Number sentence $2 \times 8 = 16, 16 + 6 = 22$

Answer **22** legs

Draw or show your work

Check-In: Multiplication

Show what you know. No help this time.

Mastery: 8 of 10 (80%)

Part 1 Groups of

1) 

4 groups of 2

4 × 2 = 8

2) 

3 groups of 2

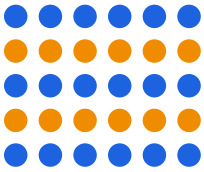
3 × 2 = 6

Part 2 Arrays

3) 

3 rows of 6

3 × 6 = 18

4) 

5 rows of 6

5 × 6 = 30

Part 3 Times 2, 5 and 10

5) 2 × 10 = 20

6) 3 × 5 = 15

7) 4 × 10 = 40

8) 2 × 5 = 10

9) 10 × 5 = 50

10) 8 × 10 = 80